

Luc Barrett

Ithaca, NY | lab392@cornell.edu | [linkedin.com/in/luc-barrett/](https://www.linkedin.com/in/luc-barrett/) | lucbarrett.info

EDUCATION

Cornell University

Ph.D. Applied Physics

Ithaca, NY

2025 — Present

University of Massachusetts Amherst

M.S. Computer Science (Bay State Fellow)

Amherst, MA

2024 — Present

- Relevant Courses: Reinforcement Learning, Machine Learning, Advanced Robot Dynamics & Control
- Final project being completed in conjunction with first-year of PhD. Expected completion Spring '26

University of Massachusetts Amherst

B.S. Physics, B.S. Computer Science, B.S. Mathematics

Amherst, MA

2020 — 2024

- GPA: 3.94/4.0 | Cum Laude, Honors Scholar with Greatest Distinction, Phi Beta Kappa
- Kandula Sastry Undergraduate Award for the outstanding physics student in the graduating class
- Thesis: “*Characterization of Drifting Charge Clusters in Liquid Xenon Produced by a Laser-Driven Photocathode for the nEXO Experiment*”, advised by Prof. Krishna Kumar and Dr. Jason Bane.
- Relevant Courses: NLP, Artificial Intelligence, Differential Geometry, Nonlinear Dynamics/Chaos, Partial Diff. Equations

EXPERIENCE

Graduate Teaching Assistant

College of Engineering, Cornell University

Aug 2025 — Present

Ithaca, NY

- Teaching assistant for AEP3330/AEP5330: Mechanics of Particles & Solid Bodies

Master's Research

College of Information & Computer Science, UMass Amherst

Sept 2024 — Present

Amherst, MA

- Trained modern reinforcement learning algorithms (e.g. PPO, DDPG) on quantum circuit compilation problems for distributed quantum computers (DQCs). Compared their performance to other graph-based methods
- Applying reinforcement learning to study optimal policies for entangling quantum memory registers under different noise models and/or assumptions

Undergraduate Research Assistant

Department of Physics, UMass Amherst

May 2021 — Aug 2024

Amherst, MA

- **nEXO Experiment - Electron Lifetime Monitoring in Liquid Xenon** (*Summer 2023*)
 - Developed several algorithms for noise reduction, including training a generative model to reconstruct a clean signal
 - Configured and calibrated signal processing chain. Developed automated pipeline for calibrating charge-sensitive preamplifier and shaper units to adjust for drifts
- **nEXO Experiment - GPU-Accelerated Optical Simulation Code** (*Summer 2024*)
 - Optimized codebase to shorten individual simulation runtime by over 10x
 - Implement test suite to ensure physical accuracy as improvements were made, and overhauled API to allow for controlled ‘sweeps’ over parameter spaces. Comparing to experimental data, this let us put constraints on some optical properties.
 - Presented at internal nEXO Light Simulation Workshop hosted at McGill University in Fall of 2024.
- **MOLLER Experiment - Path-Traced Radiation Background Simulation** (*Summer 2022*)
 - Programmed in Rust. This 3D version of the software outperformed the previous 2D version’s runtime by over 50x.
 - Improved user experience by using standardized files and providing warnings for possible geometry errors
 - Presented poster showing work at American Physical Society - Division of Nuclear Physics meeting in Fall of 2023.
- **Efficient Simulator for Gaussian Quantum States** (*Fall 2023*)
 - Wrote essential functionality for representing arbitrary n -mode gaussian states and common operations (squeeze, rotate, etc) in Julia. Presented work comparing performance over state-vector based methods.

SKILLS

- **Programming Languages:** Python, Julia, Rust, C/C++. Javascript/Typescript, Java, Kotlin
- **Technologies:** Docker, React, Git, Numpy/Scipy, Mathematica, Matlab, Labview, CAD, Tensorflow, PyTorch
- **Hardware:** Cryogenic Systems, Optics, 3D Printing, Circuit Design, Soldering

Updated Nov. 2025